

無人機輔助無線感測網路之高效資料收集軌跡規劃

摘要:

在無人機 (Unmanned Aerial Vehicle, UAV) 輔助的無線感測網路 (Wireless Sensor Networks, WSNs) 中，任務完成時間 (Mission Completion Time, MCT) 是評估系統效能的重要指標。MCT 由節點間飛行時間、通訊範圍內的資料收集時間。因此，若僅追求幾何飛行路徑最短，UAV 可能傾向於沿地面節點通訊範圍邊界通過，導致傳輸速率偏低並增加資料收集時間；相反地，若過度接近節點中心，則可能增加額外飛行距離。由此可知，縮短 MCT 需要同時考量飛行軌跡與資料收集軌跡之間的耦合關係。

為解決上述問題，本論文針對單一 UAV 輔助 WSNs 資料收集任務，提出一種結合全域幾何導引與局部聯合優化的軌跡規劃方法。首先，本研究透過二階錐規劃 (SOCP) 求解訪問所有地面節點通訊範圍的最短幾何路徑，並提出**參考路徑吸引子 (Reference-Path Attractor, RPA) 機制**。該機制沿最短幾何路徑提取吸引子作為全域方向指引，在局部優化飛行進入點與離開點時，對 UAV 的飛行航向提供幾何牽引力，有效防止路徑偏離全域最短路徑。

其次，針對資料收集過程，本論文使用 V 型收集軌跡模型，利用橢圓幾何性質精確計算滿足通訊需求的內部軌跡。當資料量可由飛行收集完成時，UAV 於飛行模式下沿 V 型軌跡收集資料；當資料需求超過飛行收集能力時，則加入懸停時間以滿足節點資料量需求。

最後，針對通訊範圍重疊的節點，本研究提出**邊界重疊策略 (Boundary-Overlap Strategy, BO)**。不同於將進出點限制於通訊圓邊界的作法，BO 允許 UAV 在相鄰節點通訊範圍的重疊區域內選擇彈性切換點，並將連續重疊節點視為一組局部聯合軌跡規劃問題。透過候選進入點、重疊切換點與離開點所構成的候選點序列，BO 可在連續重疊節點群組內搜尋較短的服務軌跡，以減少不必要的折返與邊界切換造成的飛行距離。

模擬結果顯示，在不同資料需求量與節點分佈情境下，本文方法能有效利用全域參考路徑與通訊重疊區域，減少飛行距離並縮短整體 MCT。相較於僅考慮幾何最短路徑或以固定邊界點進行任務切換的方法，所提出之 RPA 與 BO 結合的軌跡規劃方法能更有效平衡飛行時間與資料收集時間，提升 UAV 輔助 WSNs 資料收集任務的執行效率。

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Introduction:

With the rapid development of the Internet of Things (IoT) and wireless communication technologies, wireless sensor networks (WSNs) have been widely applied in environmental monitoring, smart agriculture, post-disaster detection, military reconnaissance, and other fields [1], [2], [13]. In these applications, ground nodes (GNs) are mainly responsible for collecting environmental parameters, such as temperature, humidity, air quality, and water quality. In such IoT applications, each GN usually generates lightweight data packets [14], [16]. However, due to limited battery capacity and hardware cost, GNs are often unable to support long-distance wireless transmission. Therefore, transmitting sensing data back to a data center remains a major challenge [3].

Unmanned aerial vehicles (UAVs) have become a feasible solution to this problem because of their high mobility, flexible deployment, and ability to establish favorable air-to-ground (A2G) line-of-sight (LoS) communication links [4], [20]. A UAV can serve as an aerial mobile data collector by actively approaching GNs and collecting data within their communication regions. This approach not only reduces the transmission power consumption of GNs and extends the lifetime of sensing devices, but also alleviates communication interruptions caused by terrain blockage [5], [18].

In UAV-assisted data collection tasks, mission completion time (MCT) is a critical performance metric [11]. MCT refers to the total time required for the UAV to depart from the data center, complete data collection from all target GNs, and return to the data center. For time-sensitive IoT applications, such as real-time forest fire monitoring and border intrusion detection, reducing MCT allows the decision center to obtain on-site information more quickly and respond in a timely manner [6], [7]. A shorter MCT also enables the UAV to serve more GNs within its limited battery [8].

Early studies often approximated MCT by the UAV flight time and simplified the problem into a flight distance reduction problem [3]. However, when communication models and data requirements are considered, a shortest geometric path does not necessarily lead to a shorter MCT [9], [10]. According to the Shannon formula, the wireless transmission rate increases with the signal-to-noise ratio (SNR), while the SNR decreases as the communication distance between the UAV and a GN increases [15]. If the UAV passes near the boundary of a GN communication region to reduce flight distance, the flight time can be shortened, but the lower transmission rate can increase the data collection time or hovering time [9]. In contrast, if the UAV moves excessively close to the GN center to improve the transmission rate, additional flight distance and flight time can be introduced. Therefore, reducing MCT requires a proper tradeoff between flight distance and data collection time.

Since the UAV only needs to enter the communication region of each GN to collect data and does not have to fly directly above every GN, this task can be regarded as a path planning

problem with neighborhoods, namely the Traveling Salesman Problem with Neighborhoods (TSPN) [17]. However, unlike conventional TSPN, which mainly focuses on geometric path length, UAV-assisted data collection is also affected by communication distance, transmission rate, and data collection time. Therefore, the UAV needs to determine not only the visiting path among communication regions, but also the entry position, exit position, movement depth, and data collection trajectory within each communication region. Existing studies have investigated UAV-assisted WSN data collection from different perspectives. Some studies reduce the geometric path length through GN communication regions [17], while other studies consider flying mode, hovering mode, or V-shaped collection trajectories [7], [21], [22]. These studies show that MCT is affected by both flight time and data collection time. However, two issues still require further consideration. First, if global geometric direction information is not considered, local trajectory adjustment can deviate from the overall shorter flight path. Second, when adjacent GN communication regions overlap, restricting service switching to the boundary of a single communication circle can cause redundant boundary transitions and unnecessary back-and-forth movement.

Based on the above observations, this thesis focuses on a single-UAV-assisted WSN data collection task and proposes a trajectory planning method that considers both flight path and data collection efficiency. The proposed method combines global geometric information and a local data collection strategy to reduce the overall MCT of the UAV data collection task. The related mechanisms and implementation details are presented in the following chapters. The main contributions of this thesis are summarized as follows.

1. A trajectory planning method is developed to reduce the MCT of single-UAV-assisted WSN data collection by considering both the inter-region flight path and the intra-region data collection trajectory.
2. A global-direction-guided local trajectory adjustment method is introduced. The method enables the UAV to adjust the data collection trajectory within each GN while remaining close to the shortest geometric path, thereby reducing unnecessary flight distance.
3. A flexible service switching strategy is developed for consecutive GNs with overlapping communication ranges. Instead of restricting service switching to the boundary of an individual communication circle, the proposed strategy allows the UAV to utilize the shared coverage region between neighboring GNs as a switching area, thereby reducing redundant boundary transitions and unnecessary back-and-forth movement.
4. Simulation results show that the proposed method reduces the overall MCT compared with the benchmark methods under different data requirements and GN distributions. The results further indicate that global geometric guidance and overlap-

aware service switching contribute to reducing both flight distance and mission completion time.

The remainder of this thesis is organized as follows. Chapter 2 reviews related studies on trajectory planning for UAV-assisted WSN data collection. Chapter 3 introduces the system model, including the communication model and the UAV flight model. Chapter 4 describes the proposed trajectory planning method. Chapter 5 presents the simulation setup, benchmark methods, and performance evaluation. Finally, Chapter 6 concludes this thesis and discusses possible future research directions.

Related Work:

在 UAV 輔助 WSNs 資料收集任務中，MCT 不僅受到 UAV 飛行距離影響，也受到通訊範圍內資料收集方式的影響。因此，相關研究可依其主要設計重點分為兩類：一類著重於幾何軌跡規劃，將地面節點的通訊範圍視為 UAV 可拜訪區域，藉由縮短通過所有通訊區域的飛行路徑來降低飛行時間；另一類則考量 UAV 在通訊範圍內的資料收集過程，使 UAV 可依據資料需求量與通訊條件調整其收集軌跡。前者有助於提供較短的幾何參考路徑，後者則能更節省資料收集時間。

在幾何路徑規劃方面，Luo 等人 [17] 將 UAV 資料收集路徑規劃建模為帶鄰域的旅行推銷員問題 (Traveling Salesman Problem with Neighborhoods, TSPN)，並指出此問題屬於 NP-hard。該研究針對不同場景提出近似演算法，使 UAV 能在能量與時間限制下，找到可滿足所有節點資料收集需求的飛行軌跡。此類研究強調，在幾何路徑規劃中納入感測節點通訊範圍具有重要性，因為 UAV 不必固定飛至節點中心即可完成資料接收。

收集點位置對任務時間影響在 [10] 中被進一步討論。該研究指出，UAV 的資料收集點不應單純由最短飛行距離或最佳通訊品質單獨決定。若 UAV 飛至節點正上方，雖然能取得較佳通訊品質，但可能造成較長飛行距離；相反地，若 UAV 僅在通訊範圍邊界附近收集資料，雖然可縮短飛行路徑，卻可能因 UAV 與節點距離較遠而降低傳輸速率，進而增加資料收集時間。

Li 等人 [26] 針對單一 UAV 輔助資料收集問題，同時考量軌跡、高度、速度與鏈路排程對任務時間的影響。該研究指出，當 UAV 位於地面使用者的通訊範圍內即可支援資料傳輸時，UAV 不必飛至每個地面使用者正上方，而可在通訊區域內選擇適當的通過位置。基於此，[26] 將原始任務時間問題轉換為軌跡長度問題，並在給定拜訪順序後，透過凸優化求解 UAV 進入與離開各通訊範圍的位置，以縮短通過所有通訊範圍的幾何路徑長度。此方法的重要貢獻在於說明 UAV 的服務位置不應固定於節點中心，而應作為通訊範圍內的可調整變數。然而，該方法主要仍以幾何路徑長度為設計目標，若 UAV 傾向於沿通訊範圍邊界通過，可能因距離節點較遠而降低傳輸速率，進而增加資料收集時間。

除了幾何路徑規劃外，部分研究進一步討論 UAV 飛行與通訊傳輸之間的相互影響。Gong 等人 [9] 研究在滿足資料收集需求與感測器能量限制下如何縮短 UAV 飛行時間。該研究指出，單純縮短飛行距離可能使 UAV 遠離感測節點，導致傳輸速率下降，反而增加完成資料接收所需的時間。為此，該研究利用注水演算法與動態規劃聯合調整 UAV 飛行速度與感測器傳輸功率，使飛行策略與通訊條件能共同配合。此研究說明，UAV 資料收集任務中的飛行距離、飛行速度與通訊速率並非彼此獨立，而應在同一問題中共同考量。

Wang 等人 [21] 則引入飛行模式 (Flying Mode, FM) 與懸停模式 (Hovering Mode, HM) 的資料收集概念。在 FM 中，UAV 可於飛行過程中持續接收資料，適用於資料需求量較小或通訊條件較佳的情況；在 HM 中，UAV 於特定位置懸停並進行資料

收集，適用於資料需求量較大或通訊條件較差的情況。FM/HM 的區分使 UAV 可依據節點狀態與資料需求量調整收集方式，避免將所有節點皆以固定模式服務。

在使用者排程與 UAV 軌跡設計方面，[12] 研究單一 UAV 在多個感測節點間進行資料收集時的完成時間問題。該研究證明，在特定條件下，較合適的 UAV 軌跡具有連續懸停-飛行 (Successive Hover-Fly, SHF) 結構，即 UAV 在部分位置懸停以收集資料，並在這些懸停位置之間以最大速度飛行。此外，該研究也指出基於區段 (Segment-based) 的服務排程具有重要性，亦即 UAV 在某一飛行或懸停區段中主要服務特定感測節點，而非頻繁切換服務對象。

Li 等人 [7] 研究說明了資料收集軌跡本身對 MCT 的影響，並提出 V-shaped collection trajectory。若無人機能飛入通訊範圍內部，雖然增加了部分飛行距離，但因提升了信噪比與傳輸速率，進而縮短資料收集時間，降低總任務時間。此軌跡將 UAV 飛入通訊範圍與飛離通訊範圍的過程皆納入資料收集，使 UAV 不僅在接近節點時接收資料，也能在離開節點前持續完成資料傳輸。此外，該研究亦指出相鄰節點之間的服务軌跡具有耦合性。前一節點會影響 UAV 服務下一節點時的起始位置，因此單一節點若僅獨立決定，可能對後續節點造成額外飛行距離。換言之，局部降低某一節點的資料收集時間，不一定能縮短整體 MCT；UAV 仍須考量相鄰節點之間的連續飛行方向與服務銜接。

綜合上述研究可以看出，UAV 輔助資料收集軌跡規劃主要涉及兩個相互關聯的設計面向。第一，幾何路徑規劃方法，縮短 UAV 通過多個通訊區域的飛行距離 [17], [26]。第二，資料收集軌跡設計則進一步指出，UAV 可依據資料需求量與通訊條件適度深入通訊範圍內部，以改善傳輸速率並減少資料收集時間 [7], [10]。此外，既有研究亦從飛行速度與傳輸功率控制、連續懸停-飛行結構，以及飛行或懸停模式切換等角度，說明 UAV 飛行行為與通訊條件之間具有密切關聯 [9], [12], [21]。然而，若軌跡規劃主要依據幾何路徑長度，可能無法充分反映 UAV 與地面節點距離變化對資料收集時間的影響；相反地，若局部資料收集軌跡缺乏全域幾何方向參考，則可能使局部服務軌跡逐步偏離較短的整體飛行方向。另一方面，當相鄰或連續多個節點的通訊範圍發生重疊時，重疊區域提供了額外的服務切換彈性；如何有效利用此幾何特性以減少不必要的邊界切換與折返，仍有進一步探討空間。

基於上述研究脈絡，本文聚焦於單一 UAV 輔助 WSNs 資料收集任務，探討如何同時考量全域飛行方向、通訊範圍內資料收集軌跡，以及相鄰節點間的服務銜接。既有研究已分別說明幾何路徑縮短與深入通訊範圍內部收集資料的重要性，但在局部資料收集軌跡調整過程中，如何維持與整體較短飛行方向的一致性，仍有進一步探討空間。此外，當相鄰或連續多個節點的通訊範圍發生重疊時，重疊區域可提供額外的服務切換彈性，若能適當利用此幾何特性，將有助於減少不必要的邊界切換與折返飛行。基於此，本文提出結合 Reference-Path Attractor (RPA) 機制與 Boundary-Overlap Strategy (BO) 的軌跡規劃方法，以在保留通訊範圍內資料收集效益的同時，降低局部軌跡偏離全域飛行方向的可能性，並進一步利用重疊區域中的彈性切換位置，以減

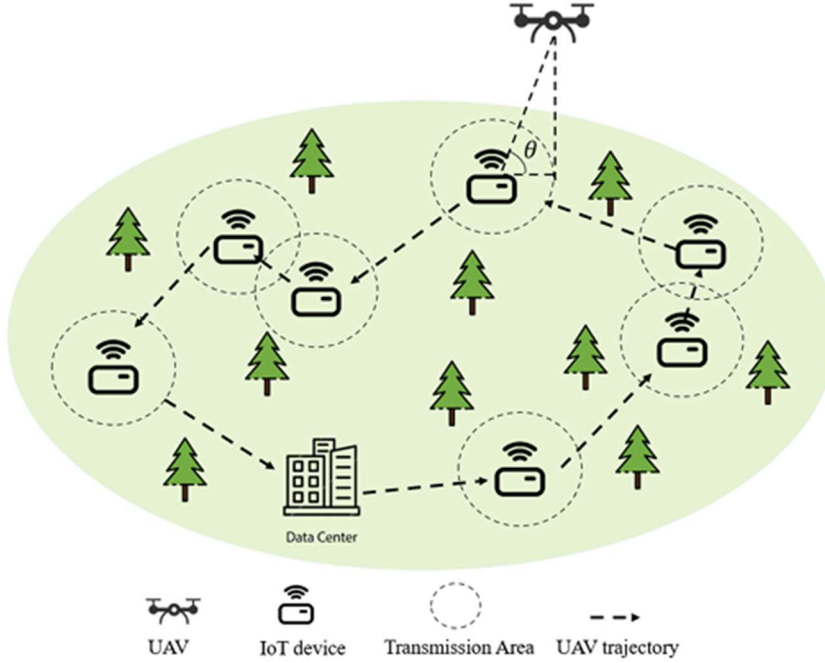
少整體 MCT。

System Model:

3.1 System Model

As shown in Fig. 1, this thesis considers a UAV-assisted wireless sensor network in which a single UAV is employed to collect data from multiple ground nodes (GNs). The UAV departs from the data center, visits the communication regions of the GNs to collect their data, and returns to the data center after completing the data collection task. Let $\mathcal{N} = g_1, g_2, \dots, g_N$ denote the set of GNs, where N is the number of GNs. A three-dimensional Cartesian coordinate system is adopted, where the GNs are located on the ground plane and the UAV flies at a fixed altitude H . The elevation angle between the UAV and a GN is denoted by θ , and the maximum flying speed of the UAV is denoted by $v_{\max}$.

During the data collection process, the UAV may operate in flying mode (FM) or hovering mode (HM). In FM, the UAV collects data while moving within the communication region of a GN. In HM, if the data collected along the flying trajectory is insufficient, the UAV hovers directly above the corresponding GN for additional data collection. Except for hovering-based data collection, the UAV flies at the maximum speed $v_{\max}$.



圖一：無人機收集資料示意圖

(1) Transmission Model:

A delay-tolerant UAV-assisted WSN is considered, in which each ground node g_i stores sensing data and uploads the data to the UAV when the UAV enters its communication region.

Each ground node is equipped with a single omnidirectional antenna and transmits with a fixed power P_t over a channel bandwidth B . The uplink transmission rate from GN g_i to the UAV at time t is expressed as

$$R_i(t) = B \log_2(1 + \gamma_i(t))$$

in which $\gamma_i(t)$ denotes the received signal-to-noise ratio (SNR) from GN g_i to the UAV at time t . The SNR is given by

$$\gamma_i(t) = \frac{P_t}{\sigma^2 L_{dB_i}(t)}$$

where σ^2 denotes the noise power at the UAV receiver, and $L_{dB_i}(t)$ represents the air-to-ground path loss between GN g_i and the UAV at time t . Similar to [26], successful data reception is guaranteed only when the received SNR is no less than a predefined threshold $\bar{\gamma}$, i.e., $\gamma_i(t) \geq \bar{\gamma}$. To avoid transmission interference among ground nodes, only the ground node being served is allowed to transmit data to the UAV during its corresponding service interval.

(2) Channel Mode:

Considering that the mission completion time generally exceeds the channel coherence time and that the UAV position changes during the mission, a statistical channel model is adopted. The path loss from GN g_i to UAV can be expressed as

$$L_{dB_i}(t) = \left(\frac{4\pi f_c}{c}\right)^2 l_i^\eta [P_{\text{LoS}_i}(t)(\xi_{\text{LoS}} - \xi_{\text{NLoS}}) + \xi_{\text{NLoS}}]$$

where f_c denotes the carrier frequency, c is the speed of light, and η is the path-loss exponent. The parameters ξ_{LoS} and ξ_{NLoS} represent the average additional path losses under LoS and NLoS propagation conditions, respectively. In addition, $l_i(t)$ denotes the three-dimensional distance between GN g_i and the UAV, which is given by

$$l_i(t) = \sqrt{(U_x(t) - GN_{ix})^2 + (U_y(t) - GN_{iy})^2 + H^2}$$

where (GN_{ix}, GN_{iy}) is the horizontal coordinate of GN g_i , $(U_x(t), U_y(t))$ is the horizontal coordinate of the UAV at time t , and H is the fixed flying altitude of the UAV.

The LoS probability between GN g_i and the UAV is modeled as

$$P_{\text{LoS}_i}(t) = \frac{1}{1 + \alpha \exp[-\beta(\theta_i(t) - \alpha)]}$$

where α and β are parameters determined by the propagation environment. The elevation angle from GN g_i to the UAV at time t is denoted by $\theta_i(t)$, which is defined as

$$\theta_i(t) = \frac{180}{\pi} \arcsin\left(\frac{H}{l_i(t)}\right).$$

According to this channel model, the path loss is affected by both the UAV–GN distance and the elevation angle. When the UAV moves closer to a GN, the three-dimensional distance decreases and the elevation angle increases, which generally improves the LoS probability and reduces the average path loss. Therefore, the UAV trajectory directly affects the received SNR and the corresponding data collection time.

3.2 Problem Formulation

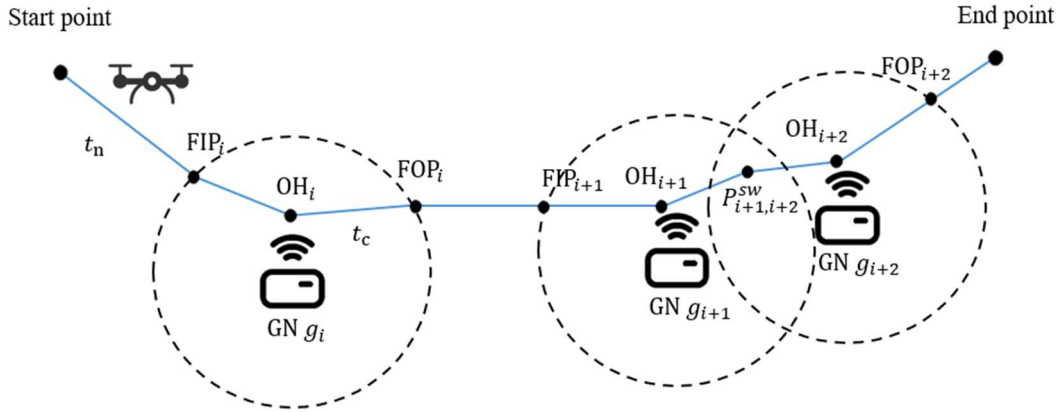


Fig. 2. UAV data collection process.

In the considered single-UAV data collection mission, all GNs must be served within one flight. The UAV starts from the data center, collects the required data from each GN, and returns to the data center. For a given transmit power P_t , the communication radius of each GN is determined by the SNR threshold $\bar{\gamma}$. Let D denote the maximum horizontal distance between the UAV and a GN such that the received SNR is equal to $\bar{\gamma}$. When the UAV is

located inside the communication region with radius D , the received SNR is no less than the required threshold, and the UAV can successfully receive data from the corresponding GN. The value of D is obtained according to the adopted LoS-based air-to-ground channel model.

The MCT of the UAV is obtained by accumulating the service time required for all GNs. For each GN, the service time consists of the flight time required to reach its communication region and the data collection time spent within that region. The index i follows the service sequence along the UAV trajectory. The GN visiting sequence is first determined by a genetic algorithm (GA), and the proposed method plans the complete UAV trajectory based on this sequence. As shown in Fig. 2, GN g_i is served independently, whereas GN g_{i+1} and GN g_{i+2} share an overlapping communication region.

When GN g_i is served independently, its service trajectory is described by three key points: the flying-in point FIP_i , the optimal hovering or turning point OH_i , and the flying-out point FOP_i . The UAV first flies from the previous service point to FIP_i , and this flight duration is denoted by t_{n_i} . After entering the communication region of GN g_i , the UAV collects data while moving from FIP_i to FOP_i through OH_i . The corresponding data collection time is denoted by t_{c_i} .

When the communication regions of two consecutive GNs do not overlap, the UAV enters and leaves the communication region of each GN through its own FIP and FOP, respectively. In this case, FIP_i and FOP_i are located on the boundary of the communication region with radius D , and the data collection trajectory of GN g_i is represented as

$$FIP_i \rightarrow OH_i \rightarrow FOP_i.$$

When the communication regions of GN g_{i+1} and GN g_{i+2} overlap, the service transition between the two GNs is represented by a common point selected from the overlapping region. Therefore, the FOP of GN g_{i+1} and the FIP of GN g_{i+2} are not forced to lie on the boundaries of their respective communication regions. To describe this service transition, this thesis introduces an overlap switching point, denoted by $P_{i+1,i+2}^{sw}$. The switching point satisfies

$$P_{i+1,i+2}^{sw} \in D_{i+1} \cap D_{i+2}$$

where D_i denotes the communication region of GN g_i . The switching point simultaneously serves as the FOP of GN g_{i+1} and the FIP of GN g_{i+2} , which is expressed as

$$FOP_{i+1} = P_{i+1,i+2}^{sw} = FIP_{i+2}.$$

Accordingly, the service trajectory across the overlapping communication regions of GN

g_{i+1} and GN g_{i+2} can be represented as

$$\text{FIP}_{i+1} \rightarrow \text{OH}_{i+1} \rightarrow P_{i+1,i+2}^{\text{sw}} \rightarrow \text{OH}_{i+2} \rightarrow \text{FOP}_{i+2}.$$

This representation indicates that the UAV completes the data collection of GN g_{i+1} before reaching the switching point, and then continues the data collection of GN g_{i+2} after passing through the same point. This setting keeps the trajectory continuous and allows the UAV to complete the service transition within the overlapping communication region, thereby avoiding unnecessary boundary departure and re-entry between consecutive GNs. For a group of consecutive overlapping GNs, the same concept is applied by defining an overlap switching point between each pair of adjacent GNs in the group.

Except for hovering-based data collection, the UAV flies at the maximum speed $v_{\max}$. Therefore, the flight time is proportional to the corresponding flight distance. For the flight segment from the previous service point to FIP_i , the flight time t_{n_i} is expressed as

$$t_{n_i} = \frac{d_{i1}}{v_{\max}}$$

where

$$d_{i1} = \|\text{FIP}_i - \text{FOP}_{i-1}\|.$$

For $i = 1$, FOP_0 denotes the data center. If GN g_i and the previously served GN overlap, FIP_i can be the corresponding overlap switching point, and the fly-in distance is adjusted accordingly. After completing the data collection task of the last GN, the UAV returns to the data center. The return time is given by

$$t^r = \frac{\|\text{DC} - \text{FOP}_N\|}{v_{\max}}$$

where DC denotes the data center.

The data collection time of GN g_i , denoted by t_{c_i} , is evaluated according to the adopted V-shaped collection trajectory. During this period, the UAV collects data while flying within the communication region. If the data collected during flight is insufficient, the UAV hovers directly above the corresponding GN to complete the remaining data collection. Let Q_i denote the required data amount of GN g_i . The accumulated received data during the collection interval must satisfy

$$\int_0^{t_{c_i}} R_i(t) dt \geq Q_i.$$

Therefore, the MCT of the UAV is expressed as

$$T = \sum_{i=1}^N (t_{c_i} + t_{n_i}) + t^r.$$

Let $\mathcal{P}$ denote the set of service points associated with the UAV trajectory, including the FIPs, OHs, FOPs, and the overlap switching points when communication regions overlap. Based on the above definitions, the overall MCT reduction problem is formulated as

$$(\mathbf{P1}): \min_{U(t), \mathcal{P}} \left[\sum_{i=1}^N (t_{c_i} + t_{n_i}) + t^r \right]$$

s.t.

$$\gamma_i(t) \geq \bar{\gamma}$$

$$\text{FIP}_i, \text{OH}_i, \text{FOP}_i \in D_i$$

$$P_{i,i+1}^{\text{sw}} \in D_i \cap D_{i+1}, i \in \mathcal{O}$$

$$\int_0^{t_{c_i}} R_i(t) dt \geq Q_i$$

Here, $\mathcal{O}$ denotes the set of consecutive GN indices whose communication regions overlap, i.e., $D_i \cap D_{i+1} \neq \emptyset$. Constraint (17a) ensures that the received SNR satisfies the required threshold during data collection. Constraint (17b) requires the key service points of each GN to lie within the corresponding communication region. Constraint (17c) defines the feasible region of the overlap switching point. Constraint (17d) guarantees that the required data amount of each GN is collected.

Problem **P1** is difficult to solve directly because the flight trajectory and the data collection trajectory are coupled. The selection of service points in $\mathcal{P}$ affects both the flight time t_{n_i} and the data collection time t_{c_i} . Moreover, when consecutive communication regions overlap, the overlap switching points affect the continuity of the service trajectory across adjacent GNs. Therefore, this thesis addresses **P1** through two trajectory planning components in the proposed solution: single-node trajectory planning and overlapping-node group trajectory planning. The RPA mechanism provides reference-path attractors for both components, whereas the BO strategy is specifically applied to overlapping-node groups to

determine feasible service switching points within overlapping communication regions.

Proposed Solution

This chapter presents the proposed trajectory planning method for reducing the mission completion time (MCT) in UAV-assisted WSNs. Since the MCT is determined by both flight time and data collection time, a trajectory with the shortest geometric length does not necessarily lead to the shortest mission duration. Therefore, the proposed method is developed based on two design considerations: reducing both the flight distance and the data collection time of each GN.

To achieve this objective, this thesis first constructs a shortest geometric reference path to provide global directional information. Based on this reference path, the Reference-Path Attractor (RPA) mechanism is used to guide the subsequent local trajectory planning process. The final UAV trajectory is then determined according to the data collection requirement of each GN and the spatial relationship among communication regions. Through this design, the UAV trajectory can remain close to the shortest geometric reference path while allowing the UAV to move deeper into the communication region when such movement can improve the transmission rate and reduce the data collection time.

Based on the spatial relationship among GNs, the proposed trajectory planning process is divided into two cases. For an independent GN whose communication region does not overlap with the communication regions of other GNs, this thesis applies single-node trajectory planning to determine the corresponding service trajectory. For consecutive GNs with overlapping communication regions, this thesis adopts the Boundary-Overlap Strategy (BO), which treats the overlapping GNs as a group and uses the flexibility provided by the overlapping regions to determine the service trajectory.

The remainder of this chapter is organized as follows. Section 5.1 introduces the construction of the shortest geometric reference path and the Reference-Path Attractor (RPA) mechanism. Section 5.2 presents the single-node trajectory planning method. Section 5.3 describes the overlapping-node group trajectory planning method based on BO.

4.1 Reference Path Construction and RPA Mechanism

The purpose of constructing a shortest geometric reference path is to provide a global geometric direction for the subsequent trajectory planning process. If the trajectory segment

associated with each GN is determined only according to the data collection time of the current GN, the selection of FIP and FOP may lack guidance from the shortest geometric reference path. As a result, the subsequent trajectory may deviate from the shortest geometric reference path, leading to an increase in flight distance. Therefore, this thesis first computes a shortest geometric reference path that passes through all GN communication regions and then extracts reference-path attractors from this path. In this thesis, an attractor is defined as a reference point extracted from the shortest geometric reference path within the communication region of a GN. The attractor is used as a directional reference when evaluating candidate FIP/FOP pairs in later trajectory planning, but the UAV is not required to pass through the attractor.

Following the service order defined in the problem formulation, the index i denotes the i -th GN visited by the UAV, and the communication region of this GN is denoted by $\mathcal{D}_i$, as defined in the system model. For the given visiting sequence, this thesis computes a shortest geometric path that passes through all communication regions $\mathcal{D}_i$. To formulate the reference path construction problem, two auxiliary reference-path points are introduced for each communication region. Let $\mathbf{r}_i^{\text{in}}$ and $\mathbf{r}_i^{\text{out}}$ denote the reference-path entry point and reference-path exit point associated with $\mathcal{D}_i$, respectively. These two points are used only for constructing the shortest geometric reference path and are not the final FIP and FOP used for data collection. The shortest geometric reference path is obtained by solving

$$\begin{aligned}
& \min_{\{\mathbf{r}_i^{\text{in}}, \mathbf{r}_i^{\text{out}}\}_{i=1}^N} \quad \|\mathbf{r}_1^{\text{in}} - \mathbf{q}_0\| + \sum_{i=1}^N \|\mathbf{r}_i^{\text{out}} - \mathbf{r}_i^{\text{in}}\| \\
& \quad + \sum_{i=1}^{N-1} \|\mathbf{r}_{i+1}^{\text{in}} - \mathbf{r}_i^{\text{out}}\| + \|\mathbf{q}_0 - \mathbf{r}_N^{\text{out}}\| \\
& \text{s. t.} \quad \mathbf{r}_i^{\text{in}} \in \mathcal{D}_i \\
& \quad \mathbf{r}_i^{\text{out}} \in \mathcal{D}_i
\end{aligned}$$

where $\mathbf{q}_0$ denotes the data center. The first term represents the distance from the data center to the reference-path entry point of the first visited GN. The second term represents the geometric segment inside each communication region. The third term represents the distance from the reference-path exit point of the current GN to the reference-path entry point of the next visited GN. The last term represents the return distance from the last communication region to the data center. Since the objective function is composed of Euclidean norm terms and the feasible communication regions are convex, (5.1) can be formulated as a SOCP problem.

The segment from $\mathbf{r}_i^{\text{in}}$ to $\mathbf{r}_i^{\text{out}}$ represents only the portion of the shortest geometric

reference path inside the communication region $\mathcal{D}_i$, rather than the actual data collection trajectory of the i -th GN. The final FIP, OH, and FOP are determined later according to the data requirement of the GN and the corresponding data collection time. Therefore, the shortest geometric reference path provides geometric reference information for the subsequent trajectory planning process.

After the shortest geometric reference path is obtained, the RPA mechanism extracts one attractor for each GN. Let the shortest geometric reference path be denoted by $\mathcal{P}_{\text{ref}}$. The path $\mathcal{P}_{\text{ref}}$ is scanned according to the traveling direction from the data center. For the i -th GN, the candidate attractor points are the points on $\mathcal{P}_{\text{ref}}$ that lie within the communication region $\mathcal{D}_i$. Let $\ell(\mathbf{p})$ denote the cumulative path length from $\mathbf{q}_0$ to point $\mathbf{p}$ along $\mathcal{P}_{\text{ref}}$. The attractor of the i -th GN is defined as

$$\mathbf{a}_i = \arg \min_{\mathbf{p} \in \mathcal{P}_{\text{ref}} \cap \mathcal{D}_i} \ell(\mathbf{p}).$$

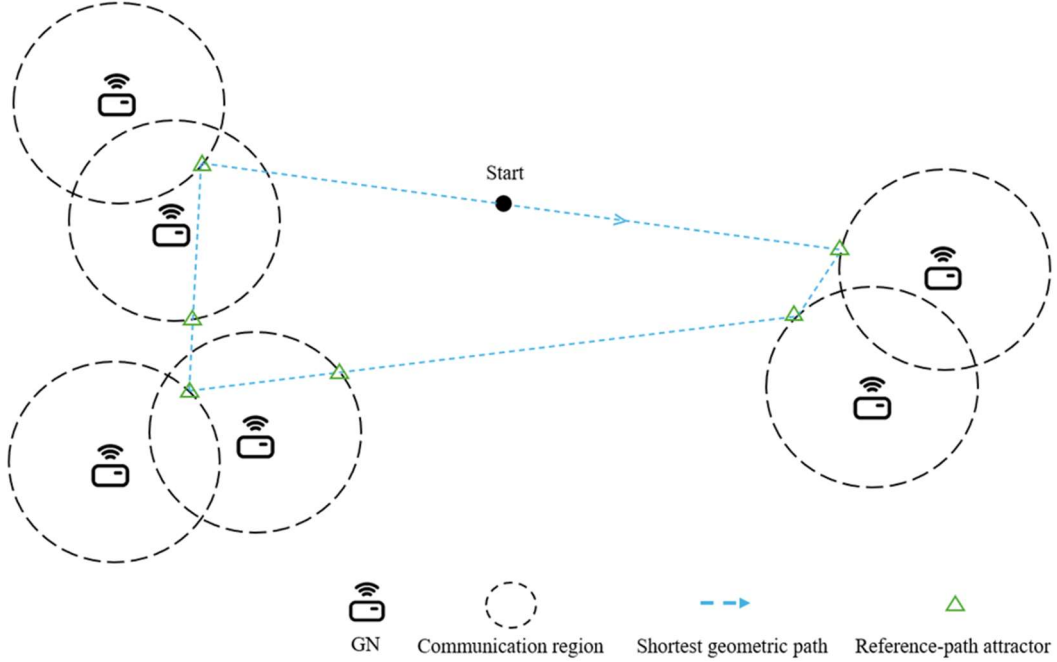


Fig. 3. Illustration of reference-path attractors.

As shown in Fig. 3, the shortest geometric path passes through the communication regions of the GNs according to the visiting direction. The triangular markers indicate the reference-path attractors extracted from this path. Each attractor is selected from the portion of the shortest geometric path that lies within the corresponding communication region.

The definition in (5.2) selects the first reference-path point that lies within the

communication region of the i -th GN along the traveling direction. This selection allows the reference path to enter the communication region of the i -th GN as early as possible, thereby preserving a longer available data collection interval along the reference path. Therefore, $\mathbf{a}_i$ represents the reference-path entry point associated with the i -th GN. The terminal attractor is set as the data center, $\mathbf{a}_{N+1} = \mathbf{q}_0$.

The influence of the attractor can be explained by decomposing the flight-time term associated with the i -th GN. Let $\mathbf{P}_i^{\text{pre}}$ denote the point immediately before serving the i -th GN. The point $\mathbf{P}_i^{\text{pre}}$ can be the FOP of the previous GN, an overlap switching point, or the data center when $i = 1$. For a candidate FIP $_i$, the fly-in time is expressed as

$$t_{n_i}^{\text{in}} = \frac{\|\text{FIP}_i - \mathbf{P}_i^{\text{pre}}\|}{v_{\max}}.$$

After the data collection process of the i -th GN is completed, the outgoing flight direction is evaluated with respect to the next attractor $\mathbf{a}_{i+1}$. The RPA-guided fly-out time is expressed as

$$t_{n_i}^{\text{out}} = \frac{\|\mathbf{a}_{i+1} - \text{FOP}_i\|}{v_{\max}}. \quad (5.5)$$

Accordingly, the flight-time term considered in the RPA-guided trajectory planning process is written as

$$t_{n_i}^{\text{RPA}} = t_{n_i}^{\text{in}} + t_{n_i}^{\text{out}}. \quad (5.6)$$

The fly-out term $t_{n_i}^{\text{out}}$ shows how the next attractor affects the evaluation of the current FOP. When different FIP/FOP pairs are considered in the trajectory evaluation process, the selected FOP determines the flight time toward the next attractor. Therefore, the RPA mechanism affects the selection of the FIP/FOP pair through the RPA-guided flight-time term.

A candidate FIP/FOP pair is evaluated not only according to the data collection time of the current GN, but also according to the flight time toward the reference direction derived from the shortest geometric reference path. The UAV trajectory is not constrained to pass through the attractor. The complete candidate evaluation rule is described in Section 5.2.

Algorithm 1 summarizes the procedure for constructing the shortest geometric reference path and extracting the corresponding reference-path attractors. Based on the extracted attractors and the RPA-guided flight-time term, Section 5.2 presents the single-node trajectory planning method, including the generation of candidate FIP/FOP pairs, the determination of OH, and the evaluation of t_{c_i} . Section 5.3 further describes how the trajectory planning process is modified when consecutive GNs have overlapping

communication regions.

===== Pseudo code 1 =====

```

\begin{algorithm}[t]
\caption{Reference Path Construction and RPA Mechanism}
\label{alg:rpa}
\begin{algorithmic}[1]
\Require Communication regions  $\{\mathcal{D}_i\}_{i=1}^N$  in the visiting order;
data center  $\mathbf{q}_0$ ; UAV maximum speed  $v_{\max}$ 
\Ensure Shortest geometric reference path  $\mathcal{P}_{\mathrm{ref}}$ ; reference-path
attractors  $\{a_i\}_{i=1}^{N+1}$ 

\State Solve the SOCP problem in (5.1) to obtain
 $\{\mathbf{r}^{\mathrm{in}}_i, \mathbf{r}^{\mathrm{out}}_i\}_{i=1}^N$ .
\State Construct the shortest geometric reference path  $\mathcal{P}_{\mathrm{ref}}$ 
according to
\[,
\mathbf{q}_0
\rightarrow
\mathbf{r}^{\mathrm{in}}_1
\rightarrow
\mathbf{r}^{\mathrm{out}}_1
\rightarrow
\cdots
\rightarrow
\mathbf{r}^{\mathrm{in}}_N
\rightarrow
\mathbf{r}^{\mathrm{out}}_N
\rightarrow
\mathbf{q}_0 .
\]

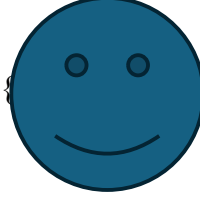
\For{$i=1$ to $N$}
\State Scan  $\mathcal{P}_{\mathrm{ref}}$  along the traveling direction from
 $\mathbf{q}_0$ .
\State Find the first point on  $\mathcal{P}_{\mathrm{ref}}$  that lies within
 $\mathcal{D}_i$ :

```

```

\l
a_i
=
\arg\min_{p\in \mathcal{P}_{\{\mathrm{ref}\}}} \mathcal{D}_i
l(p),
\l

```



where $l(p)$ denotes the cumulative path length from $\mathbf{q}_0$ to p along $\mathcal{P}_{\{\mathrm{ref}\}}$.

```
\EndFor
```

\State Set the terminal attractor as the data center:

```

\l
a_{N+1}=\mathbf{q}_0 .
\l

```

```
\State \Return  $\mathcal{P}_{\{\mathrm{ref}\}}$  and  $\{a_i\}_{i=1}^{N+1}$ .
```

```
\end{algorithmic}
```

```
\end{algorithm}
```

需要的 package :

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\usepackage{algorithm}
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\usepackage{algpseudocode}
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=====

4.2 Single-Node Trajectory Planning

This section describes the trajectory planning process for an independent GN. An independent GN is defined as a GN whose communication region does not overlap with the communication regions of other GNs. Based on the RPA-guided flight-time term defined in Section 5.1, this planning process selects a suitable FIP/FOP pair while considering the data collection time inside the communication region. The corresponding OH is determined during the evaluation of the V-shaped data collection trajectory.

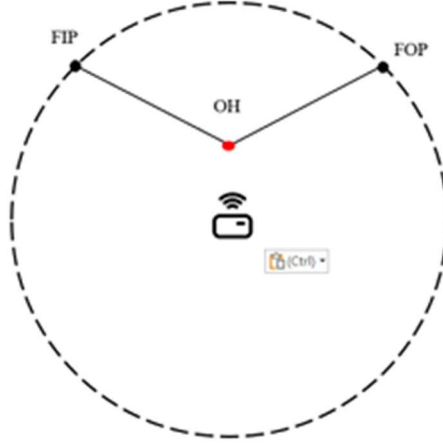


Fig. 4. Illustration of the V-shaped trajectory for single-node data collection.

As shown in Fig. 4, the UAV collects data while moving inside the communication region of the GN. Candidate FIP and FOP points are generated on the communication boundary of the GN. Let $\mathcal{F}_i^{\text{in}}$ and $\mathcal{F}_i^{\text{out}}$ denote the candidate sets of FIP and FOP, respectively. For each candidate pair $(\text{FIP}_i, \text{FOP}_i)$, the corresponding OH_i and the data collection time t_{c_i} are evaluated according to the required data amount C_i .

For each candidate pair, the corresponding V-shaped trajectory is evaluated to determine whether the required data amount can be collected during flight. If the flying collection is sufficient, FM is adopted. If the flying collection is insufficient, HM is adopted, and the UAV collects the remaining data by hovering. After the data collection time is obtained for all candidate pairs, the candidate pair is selected by considering both t_{c_i} and the RPA-guided flight-time term $t_{n_i}^{\text{RPA}}$.

For a given FIP/FOP pair, the flying distance inside the communication region is determined by the sum of the distance from FIP to OH and the distance from OH to FOP. According to the geometric property of an ellipse, all points with the same sum of distances to FIP_i and FOP_i form an elliptical curve. Therefore, different V-shaped trajectories can be evaluated by adjusting OH_i and the corresponding internal flying distance. When the required data amount can be collected during flight, a bisection search is used to determine a feasible V-shaped flying trajectory that satisfies the data collection requirement.

The amount of data collected during flight is calculated by integrating the instantaneous transmission rate along the trajectory:

$$C_i^{\text{FM}} = \int_0^{t_i^{\text{FM}}} R_i(t) dt.$$

where t_i^{FM} denotes the flying collection time of the candidate V-shaped trajectory. If $C_i^{\text{FM}} \geq C_i$, the required data amount can be collected in FM, and the data collection time is

given by $t_{c_i} = t_i^{\text{FM}}$. If $C_i^{\text{FM}} < C_i$, the flying trajectory cannot collect all required data. The UAV then operates in HM. In HM, OH is used as the hovering point, and the UAV collects the remaining data at this point. The additional hovering time is calculated as

$$t_i^{\text{H}} = \frac{C_i - C_i^{\text{FM}}}{R_i(\text{OH}_i)}.$$

Therefore, the data collection time in HM is

$$t_{c_i} = t_i^{\text{FM}} + t_i^{\text{H}}. \quad (5.11)$$

After t_{c_i} is obtained for all candidate FIP/FOP pairs, each candidate is evaluated together with the RPA-guided flight-time term. The selected pair is determined by

$$\min_{\text{FIP}_i, \text{FOP}_i} t_{c_i}(\text{FIP}_i, \text{OH}_i, \text{FOP}_i) + t_{n_i}^{\text{RPA}}. \quad (5.12)$$

In (5.12), OH_i is determined during the evaluation of t_{c_i} . Therefore, the selected pair is determined by considering both the data collection time inside the communication region and the RPA-guided flight time. After the trajectory of the current GN is determined, the resulting FOP is used as the starting point for planning the next GN. When consecutive GNs have overlapping communication regions, the corresponding overlapping-node trajectory planning method is described in Section 5.3.

4.3 Overlapping-Node Group Trajectory Planning

This section describes the trajectory planning method for GNs with overlapping communication regions. For an independent GN, the FIP and FOP candidates are generated on the communication boundary of a single GN, as described in Section 5.2. When the communication regions of adjacent GNs overlap, the overlapping region provides additional flexibility for service transition. If the UAV is still forced to leave one communication region from its boundary and enter the next communication region from another boundary point, unnecessary flight distance may be introduced.

To use the overlapping region between adjacent GNs, this thesis proposes the Boundary-Overlap Strategy (BO). BO allows the switching point $\mathbf{P}_{i,i+1}^{\text{sw}}$, defined in the system model, to be selected from the overlapping region of g_i and g_{i+1} . In this way, the UAV can connect the service trajectory of two adjacent GNs through the overlapping region, instead of performing an additional boundary-to-boundary transition.

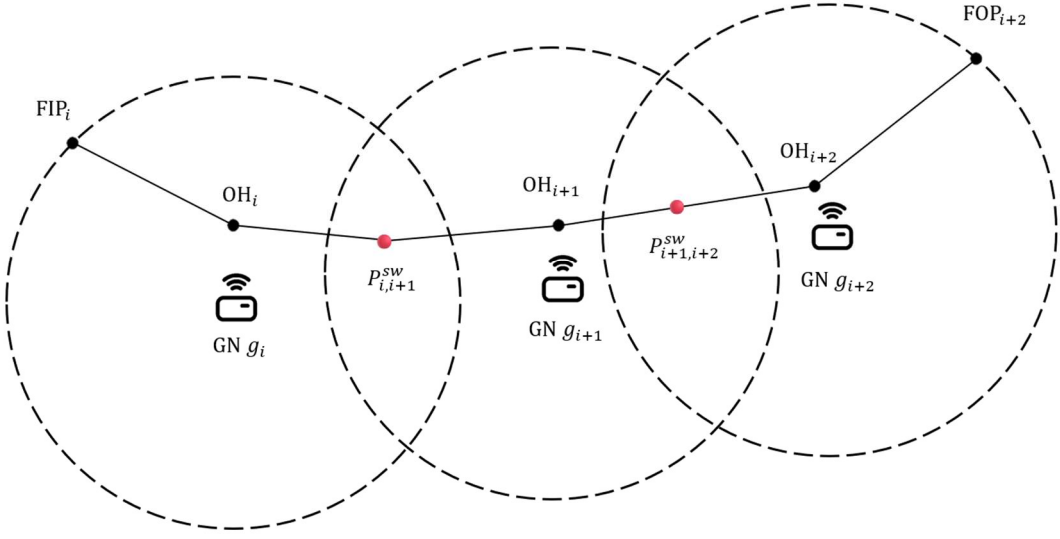


Fig. 5. Illustration of the Boundary-Overlap Strategy for overlapping GNs.

As shown in Fig. 5, multiple adjacent GNs with overlapping communication regions are considered as one group. Let

$$G_m = \{g_i, g_{i+1}, \dots, g_j\} \quad (5.13)$$

denote the m -th overlapping-node group, where g_i and g_j are the first and last GNs in the group, respectively. Here, G_m denotes a set of GNs, while g_i denotes an individual GN. For any two adjacent GNs g_k and g_{k+1} in G_m , where $k = i, \dots, j-1$, their communication regions overlap. The UAV enters the group from the FIP of g_i , passes through the switching points in the overlapping regions and leaves the group from the FOP of g_j . For example, if the group contains g_i , g_{i+1} , and g_{i+2} , the service trajectory inside the group can be represented as

$$\text{FIP}_i \rightarrow \text{OH}_i \rightarrow \mathbf{P}_{i,i+1}^{\text{sw}} \rightarrow \text{OH}_{i+1} \rightarrow \mathbf{P}_{i+1,i+2}^{\text{sw}} \rightarrow \text{OH}_{i+2} \rightarrow \text{FOP}_{i+2}. \quad (5.14)$$

In this sequence, $\mathbf{P}_{i,i+1}^{\text{sw}}$ is used as the exit point of g_i and the entry point of g_{i+1} . Similarly, $\mathbf{P}_{i+1,i+2}^{\text{sw}}$ connects the service trajectories of g_{i+1} and g_{i+2} . Therefore, the group is planned as a single service unit with one entry point and one exit point.

For each overlapping region, BO selects the switching point from a finite candidate set instead of searching over the entire continuous overlapping region. The candidate set is generated using two representative sampling directions. The first direction follows the line segment connecting the centers of the two adjacent GNs. Since different adjacent GN pairs may have different overlap depths. Therefore BO can adjust whether the service transition should occur earlier or later within the overlapping region. The second direction follows a perpendicular sampling line inside the overlapping region. This direction provides candidates with different inward depths within the overlapping region. Since different data amounts may require different collection depths, the perpendicular sampling line allows BO to adjust the switching location according to the data collection requirement. By combining these two representative directions, the T-shaped sampling method generates switching candidates for the overlapping region without searching all possible points in the continuous region.

For the m -th overlapping-node group G_m , a candidate point sequence is written as

$$\mathcal{S}_m = [\text{FIP}_i, \mathbf{P}_{i,i+1}^{\text{sw}}, \dots, \mathbf{P}_{j-1,j}^{\text{sw}}, \text{FOP}_j]. \quad (5.15)$$

The first point in $\mathcal{S}_m$ is the group entry point, and the last point is the group exit point. The intermediate points are switching points selected from the overlapping regions. Given a candidate sequence, each GN uses the previous point in the sequence as its entry point and the next point as its exit point. After a candidate sequence $\mathcal{S}_m$ is determined, the data collection time of each GN in the group is evaluated according to the V-shaped trajectory model described in Section 5.2. The collection-time term of the group is defined as the sum of the data collection times of all GNs in the group:

$$t_{c,m} = \sum_{k=i}^j t_{c_k} \quad (5.16)$$

In addition to the collection-time term, the flight time before entering and after leaving the group must also be considered. Since the overlapping-node group is served as one

continuous trajectory segment, the group has one entry point and one exit point. The flight-time term of the group is expressed as

$$t_{n,m} = \frac{\| \text{FIP}_i - \text{FOP}_{i-1} \|}{v_{\max}} + \frac{\| a_{j+1} - \text{FOP}_j \|}{v_{\max}}. \quad (5.17)$$

Therefore, each candidate sequence is evaluated by

$$T_m = t_{c,m} + t_{n,m}. \quad (5.18)$$

In (5.18), T_m denotes the evaluation time of a candidate sequence for the m -th overlapping-node group. The term $t_{c,m}$ represents the total data collection time of the GNs in the group, while $t_{n,m}$ represents the flight time associated with entering and leaving the group. The candidate sequence with the smallest T_m is selected as the service trajectory of the group. In (5.17), the first term represents the flight time from the FOP of the previous GN to the entry point of the group, and the second term represents the flight time from the exit point of the group toward the next attractor.

Since a group may contain more than two GNs, directly enumerating all possible combinations of entry points, switching points, and exit points may increase the search cost. BO therefore organizes the candidate points into layers. The first layer contains candidate FIP points of the first GN in the group, the intermediate layers contain candidate switching points generated in the overlapping regions, and the last layer contains candidate FOP points of the last GN. A dynamic programming search is then applied over these layers to find the candidate sequence with the smallest accumulated evaluation time.

After the trajectory of the group is determined, the selected FOP_j is used as the previous service point for the next planning step. When the next GN is independent, the single-node trajectory planning method in Section 5.2 is applied. When another overlapping-node group is encountered, BO is applied again. The detailed procedure of BO is summarized in Algorithm 2.

===== Algorithm 2 =====

\begin{algorithm}[t]

\caption{Boundary-Overlap Strategy for an Overlapping-Node Group}

```

\label{alg:bo}

\begin{algorithmic}[1]

\Require Overlapping-node group  $G_m = \{g_i, g_{i+1}, \dots, g_j\}$ ; communication
regions  $\{\mathcal{D}_k\}_{k=i}^j$ ; FOP of the previous GN  $\mathrm{FOP}_{i-1}$ ;
next attractor  $a_{j+1}$ ; UAV maximum speed  $v_{\max}$ 

\Ensure Selected candidate sequence  $S_m$  and evaluation time  $T_m$ 

\State Generate the candidate FIP set of the first GN  $g_i$ .

\State Set the candidate FIP set as the first layer  $\mathcal{L}_0$ .

\For{$k=i$ to $j-1$}

    \State Generate candidate switching points in  $\mathcal{D}_k \cap \mathcal{D}_{k+1}$  using the T-shaped sampling method.

    \State Set the generated switching candidates as layer  $\mathcal{L}_{k-i+1}$ .

\EndFor

\State Generate the candidate FOP set of the last GN  $g_j$ .

\State Set the candidate FOP set as the last layer  $\mathcal{L}_{j-i+1}$ .

\If{$g_i$ is the first GN in the visiting order}

    \State Set  $\mathrm{FOP}_{i-1} \leftarrow \mathbf{q}_0$ .

\EndIf

\ForAll{$p \in \mathcal{L}_0$}

```

$\Phi_0(p) \leftarrow \frac{\left| p - \text{FOP}_{i-1} \right|}{v_{\max}}.$

$\text{Pred}_0(p) \leftarrow \emptyset.$

EndFor

For $r=0$ to $j-i$

State Set the current GN index as $k=i+r$.

ForAll $q \in \mathcal{L}_{r+1}$

State Compute the minimum accumulated evaluation time:

[

$\Phi_{r+1}(q)$

=

$\min_{p \in \mathcal{L}_r}$

[

$\Phi_r(p)$

+

$t_{c_k}(\text{p}, \text{OH}_k, q)$

right].

]

State Store the predecessor point:

[

$\text{Pred}_{r+1}(q)$

=

$\arg\min_{p \in \mathcal{L}_r}$

```

\left[
\Phi_r(p)
+
t_{c_k}\left(p,\mathrm{OH}_{k,q}\right)
\right].
\]

\EndFor

\EndFor

```

\State Select the final point in the last layer:

```

\l[
p^{\mathrm{end}}
=
\arg\min_{p\in \mathcal{L}_{j-i+1}}
\left[
\Phi_{j-i+1}(p)
+
\frac{\left|a_{j+1}-p\right|}{v_{\max}}
\right].
\]

```

\State Compute the evaluation time:

```

\l[

```

T_m

=

$\Phi_{j-i+1}(p^{\mathrm{end}})$

+

$\frac{\left|a_{j+1}-p^{\mathrm{end}}\right|}{v_{\max}}.$

]

State Backtrack from p^{end} using the stored predecessors to obtain the selected candidate sequence S_m .

State Return S_m and T_m .

end{algorithmic}

end{algorithm}

=====

Performance Evaluation:

This chapter evaluates the performance of the proposed Reference-Path Attractor and Boundary-Overlap (RPA-BO) scheme for UAV-assisted data collection in wireless sensor networks. The main objective of the evaluation is to examine whether the proposed scheme can reduce the mission completion time (MCT) under different data sizes. In addition to MCT, the total flight distance is also evaluated, since the flight distance directly affects the UAV flight time and also reflects how different trajectory planning schemes guide the UAV through the GN communication regions.

The performance of the proposed RPA-BO scheme is compared with two baseline schemes. The first baseline is the Convex shortest path, which determines the shortest geometric trajectory through the GN communication regions based on the obtained GN visiting sequence. This baseline represents a geometric trajectory planning approach that mainly reduces the flight path length. To ensure a fair comparison, the same GN visiting sequence is applied to all compared schemes. The second baseline is the V-shaped scheme, which adopts a V-shaped data collection trajectory for each GN and considers the data collection process inside the communication region. The proposed RPA-BO scheme further uses reference-path attractors to provide global geometric guidance and applies the Boundary-Overlap Strategy to enable flexible service switching in overlapping GN communication regions.

5.1 Simulation Setup

The simulation area is set to 2000×2000 m². A total of 20 GNs are randomly and uniformly deployed in the simulation area, and a single UAV is used to collect data from all GNs. The data center is located at (1000, 1000) m. The UAV departs from the data center, visits all GNs according to the generated visiting order, collects the required data from each GN, and returns to the data center after completing the mission. For each data size, 200 independent simulation runs are conducted with different random GN deployments. The reported MCT and flight distance results are the average values over these 200 runs. The required data amount of each GN is denoted by C . The data amount is set to $C \in \{1, 2, 4, 8, 16, 32\}$ MB. Table 6.1 lists the main simulation parameters used in the performance evaluation.

Table 5.1 Simulation Parameters

Parameter	Value
Simulation area	2000×2000 m ²
Data center position	(1000,

Parameter	Value
	1000) m
UAV altitude	50 m
UAV maximum speed	20 m/s
GN transmit power	-40 dBm
Carrier frequency	2 GHz
Bandwidth	2 MHz
Noise power spectral density	-174 dBm/Hz
SNR threshold	2.6 dB
Path-loss exponent	1.3165
LoS additional loss	3 dB
NLoS additional loss	13 dB
LoS probability parameters	$\alpha = 11.95$, $\beta = 0.14$

Three schemes are compared in the simulation. The first scheme is the Convex shortest path, which computes the shortest geometric path passing through the communication region of each GN. This scheme is used as a geometric path baseline because its main purpose is to reduce the flight distance through the GN communication regions. The second scheme is the V-shaped scheme, which adopts the V-shaped data collection trajectory. This scheme considers the trajectory inside the communication region of each GN and adjusts the UAV service trajectory according to the data amount. The third scheme is the proposed RPA-BO scheme. This scheme uses reference-path attractors to provide global geometric guidance during local trajectory planning and applies the Boundary-Overlap Strategy to handle consecutive overlapping GN communication regions. When adjacent communication regions overlap, the UAV can select a switching point inside the overlapping region instead of leaving one communication boundary and entering another. For a fair comparison, the three schemes are evaluated under the same GN deployment and the same visiting order in each simulation run. Therefore, the differences in MCT and flight distance mainly result from the trajectory planning and data collection strategy adopted by each scheme.

5.2 Performance Comparison

Fig. 6 shows the initial trajectory of the UAV in the $2\text{ km} \times 2\text{ km}$ area. In this simulation scenario, 20 GNs are randomly and uniformly deployed in the area. By calculation, the communication radius (D) is approximately equal to 200 m. The UAV directly visits the center of each GN according to the visiting order. This trajectory is used to illustrate the GN distribution and the visiting order of the UAV before applying the proposed trajectory design.

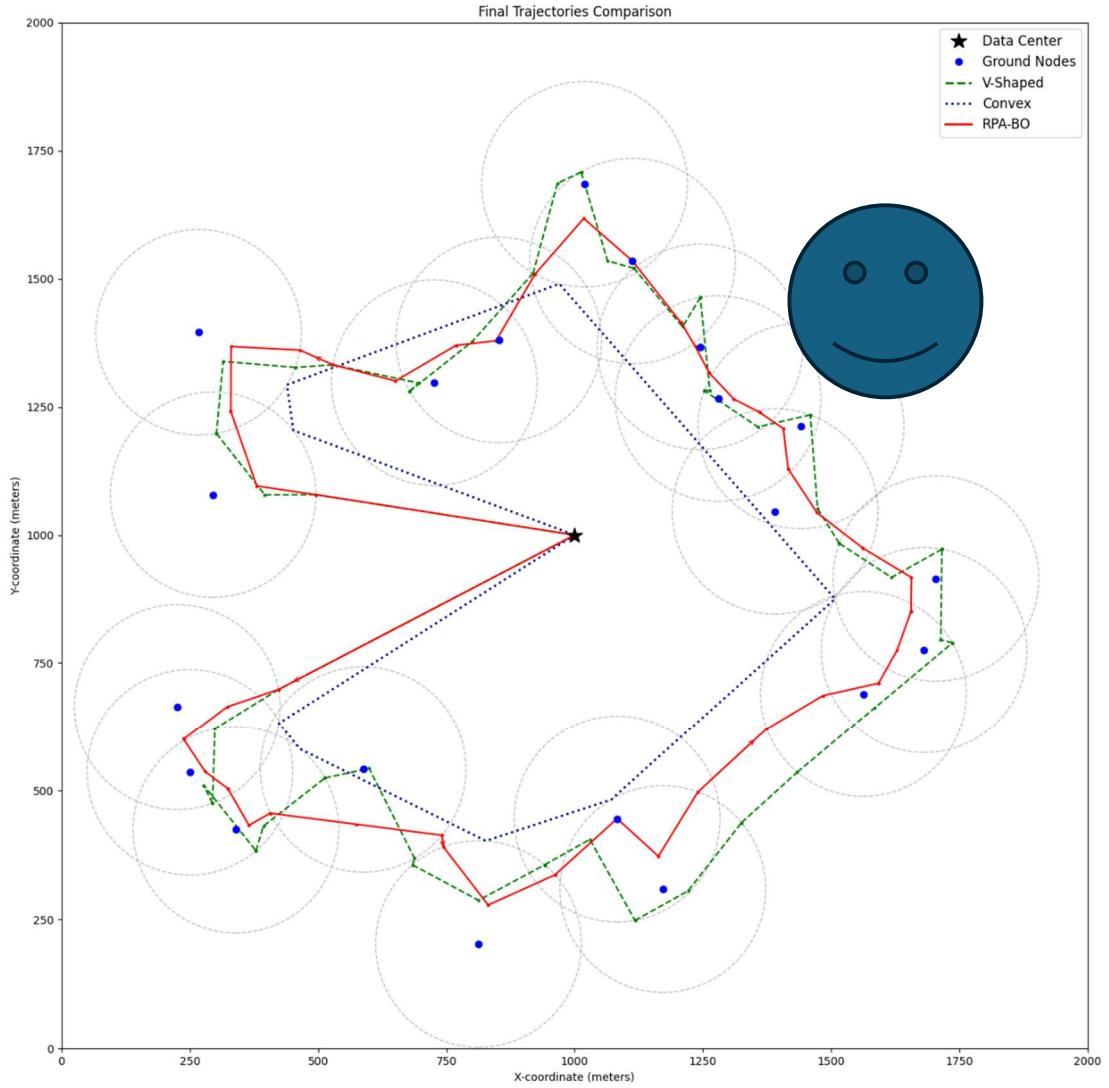


Fig. 6. Initial trajectory of UAV in $2\text{ km} \times 2\text{ km}$ area.

Fig. 7 compares the final trajectories of the proposed RPA-BO scheme, Convex shortest path, and V-shaped scheme under the 8 MB data size. In this case, the Convex shortest path has the shortest flight distance of 4407.78 m, but its MCT is 488.59 s. This result indicates that a shorter geometric flight distance does not necessarily lead to a shorter MCT, because the UAV may collect data near the communication boundary, where the transmission rate is relatively low. The V-shaped scheme reduces the MCT to 341.12 s by moving the UAV deeper into the GN communication region, so that the UAV can collect data at a higher transmission rate, its flight distance increases to 6766.34 m. The proposed RPA-BO scheme achieves the shortest MCT of 303.40 s with a flight distance of 6053.94 m.

This improvement mainly comes from two aspects. First, unlike schemes that restrict the entry and exit points to the boundaries of communication regions, the BO strategy optimizes the service switching locations within overlapping communication regions. As a result, after completing data collection from one GN, the UAV can immediately switch to serving the next GN within the overlap area, without first leaving the communication region of the current GN and then entering that of the next GN. This switching strategy reduces unnecessary flight distance and redundant movement between neighboring communication regions. Second, the RPA mechanism provides global geometric guidance during local trajectory planning, so the UAV trajectory can remain closer to the global geometric direction while still considering the data collection requirement. Therefore, the proposed RPA-BO scheme can reduce the flight distance and shorten the mission completion time.

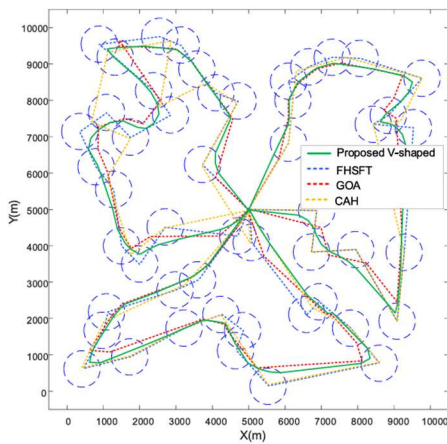


Fig. 7. Final trajectories of compared schemes under 8 MB data size.

Fig. 8 shows the average MCT of the three schemes under different data sizes. Each point represents the average result over 200 independent random GN deployments. The proposed RPA-BO scheme achieves the shortest average MCT under all tested data sizes. When $C = 1$ MB, the average MCT values of the Convex shortest path, V-shaped scheme, and proposed RPA-BO scheme are 276.14 s, 314.66 s, and 274.57 s, respectively. In this low-data-size case, the required data amount is small, so the flight distance is a major factor affecting the MCT. Nevertheless, the MCT of the proposed RPA-BO scheme is lower than that of the Convex shortest path because the proposed scheme does not merely reduce the flight distance, but also considers the data collection trajectory inside the GN communication regions. Even when the required data amount is small, selecting appropriate data collection trajectories and overlap switching points can reduce the time spent on service transitions and data collection, thereby leading to a shorter overall MCT. When $C = 4$ MB, the average MCT values of the Convex shortest path, V-shaped scheme, and proposed RPA-BO scheme are 396.95 s, 345.01 s, and 309.98 s, respectively. Compared with the Convex shortest path and the V-shaped scheme, the proposed RPA-BO scheme reduces the average MCT by 86.97 s and 35.03 s, respectively. This result shows that the proposed RPA-BO scheme can effectively reduce MCT when both flight distance and data collection time have influence on the mission. When the data size further increases, the proposed RPA-BO scheme remains effective. When $C = 32$ MB, the average MCT values of the Convex shortest path, V-shaped scheme, and proposed RPA-BO scheme are 1600.92 s, 629.42 s, and 610.41 s, respectively. Unlike approaches that mainly focus on reducing the flight distance, the proposed scheme also improves the efficiency of the data collection process. As the required data amount increases, communication efficiency becomes increasingly important. The proposed RPA-BO scheme effectively reduces data collection time and maintains the shortest MCT among all compared schemes.

The simulation results show that the proposed RPA-BO scheme achieves the shortest average MCT under all considered data sizes. Although the Convex shortest path obtains the shortest flight distance, its MCT increases rapidly as the data size increases. This result indicates that reducing flight distance alone is insufficient when the data size increases. The V-shaped scheme reduces the data collection time by moving the UAV deeper into the GN communication region when necessary, but this also increases the flight distance. The proposed RPA-BO scheme further reduces MCT by using reference-path attractors and overlap switching points. The reference-path attractors provide global geometric guidance during local trajectory planning, while the Boundary-Overlap Strategy allows the UAV to make better use of overlapping communication regions. Therefore, the proposed RPA-BO scheme can reduce flight distance and shorten the overall MCT under different data sizes.

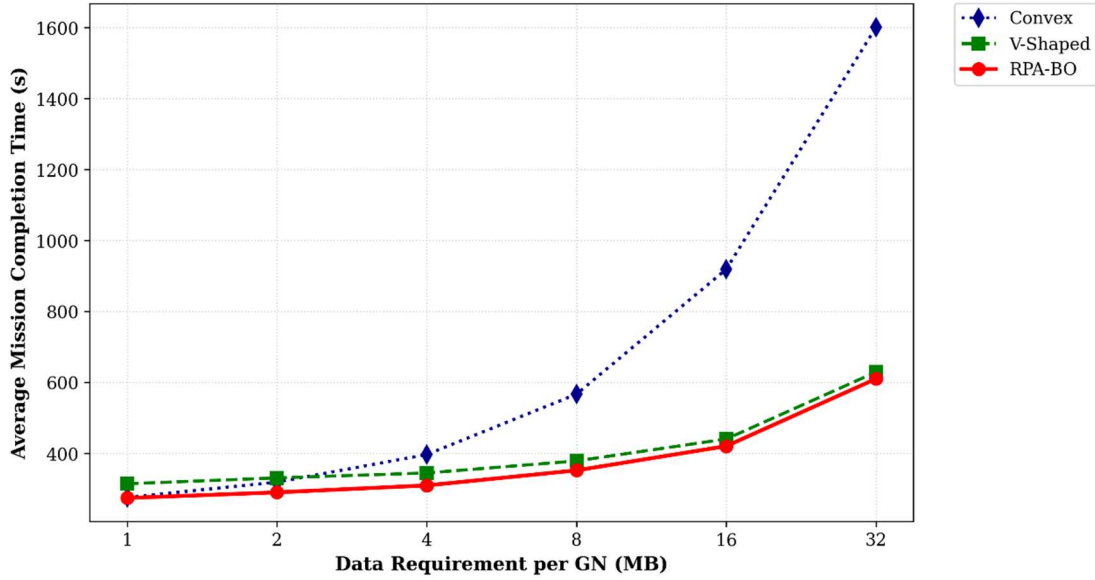


Fig. 8. Mission completion time under different data sizes.

Fig. 9 shows the average flight distance of the three schemes under different data sizes. Each point represents the average result over 200 independent random GN deployments. As shown in Fig. 8, the shortest flight distance does not correspond to the shortest MCT, especially when the required data amount becomes large. This result confirms that flight distance and data collection time should be considered together.

The V-shaped scheme has a longer flight distance because the UAV may move deeper into the GN communication regions to improve the transmission rate. In contrast, the proposed RPA-BO scheme achieves a shorter flight distance than the V-shaped scheme under all tested data sizes. When $C = 4$ MB, the average flight distances of the V-shaped scheme and the proposed RPA-BO scheme are 6894.67 m and 6191.67 m, respectively. The proposed RPA-BO scheme reduces the average flight distance by 703.00 m while also reducing the MCT, as shown in Fig. 8. When $C = 32$ MB, the average flight distances of the V-shaped scheme and the proposed RPA-BO scheme are 8534.91 m and 8170.13 m, respectively. The proposed RPA-BO scheme still maintains a shorter flight distance. This improvement is mainly attributed to the reference-path attractors and overlap switching points. The reference-path attractors guide the local trajectory planning toward the global geometric direction, while the overlap switching points allow the UAV to switch services more efficiently in overlapping communication regions. This result indicates that the proposed scheme can reduce the flight distance while maintaining the benefits of efficient data collection.

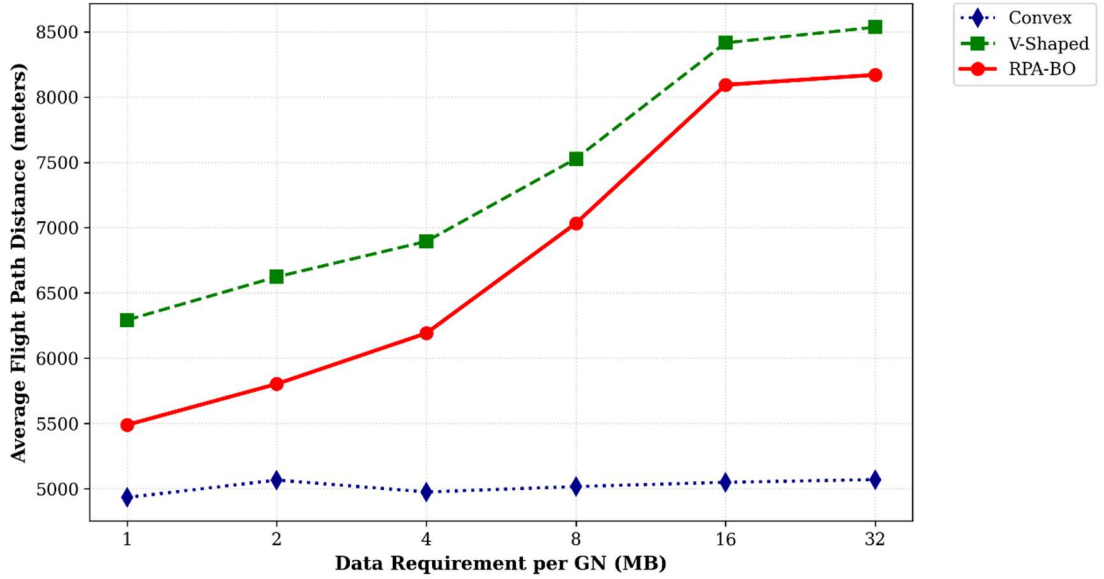


Fig. 9. Flight distance under different data sizes.

The simulation results demonstrate that reducing the flight distance alone is insufficient to reduce the MCT in UAV-assisted WSN data collection. The MCT is jointly affected by the UAV flight distance, the data collection trajectory within GN communication regions. By considering these factors together, the proposed RPA-BO

scheme achieves the shortest MCT among the compared schemes under the tested data amounts. These results confirm that integrating global geometric guidance, local data collection trajectory design, and service switching in overlapping communication regions is effective in reducing the overall MCT.

